Cluster: AIML/ANLP



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INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

Evidence-Grounded Detection of Old Videos Reposted as Recent News

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A
BE MAJOR PROJECT PROPOSAL
SUBMITTED TO THE
DEPARTMENT OF ELECTRONICS & COMPUTER ENGINEERING
IN PARTIAL FULFILLMENT FOR THE DEGREE OF
BACHELOR OF ENGINEERING IN
ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING

DEPARTMENT OF ELECTRONICS & COMPUTER ENGINEERING
LALITPUR, NEPAL

July, 2026

Abstract

Authentic videos are frequently recirculated on social media with captions that falsely present them as recent events. In Nepal, this problem is common during floods, landslides, earthquakes, protests, political incidents, fires, road accidents, and other fast-moving news situations. Unlike deepfakes, the video itself may be genuine and unedited; the misinformation arises because the original time, location, or event is replaced by a new and misleading claim.

This project proposes an evidence-grounded multimodal system for detecting whether a Nepal-related news video has previously appeared within a historical collection covering the past three years. A structured repository is constructed from publicly available and verified Nepali news sources. For each collected video, the system stores the original publication date, source, title, event category, location, transcript, on-screen text, representative keyframes, perceptual hashes, and visual embeddings.

For a query post, the pipeline extracts the caption, video keyframes, audio transcript, and on-screen text. Whisper is used for automatic speech recognition, an OCR engine is used for embedded text, and OpenCLIP-based frame embeddings together with perceptual hashes are used to retrieve earlier occurrences from a FAISS index. Retrieved candidates are reranked using visual similarity, temporal consistency, OCR overlap, transcript similarity, and metadata reliability. The system then compares the context claimed by the query post against the verified context of the best historical match.

The final output is one of three decisions: *recent or correctly contextualised*, *old video reposted as recent*, or *insufficient evidence*. Each decision is accompanied by the matched historical source and a concise explanation. The project is designed as a practical prototype for Nepali journalists, fact-checkers, researchers, and media organisations.

Keywords: multimodal misinformation, resurfaced video detection, video retrieval, Nepal news, Whisper, OCR, OpenCLIP, FAISS, evidence grounding

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Abbreviations

ASR	Automatic Speech Recognition
CLIP	Contrastive Language–Image Pretraining
CNN	Convolutional Neural Network
FAISS	Facebook AI Similarity Search
NER	Named-Entity Recognition
OCR	Optical Character Recognition
OOC	Out-of-Context
pHash	Perceptual Hash
API	Application Programming Interface
GPU	Graphics Processing Unit

1. Introduction

1.1 Background

Social media has become a major channel for consuming breaking news. Short videos are frequently shared faster than complete reports, and users often forward them without checking when or where the footage was originally recorded. This creates a particular form of misinformation in which an authentic old video is reposted with a new caption claiming that it shows a current event.

The problem is especially relevant in Nepal because natural disasters, political demonstrations, road incidents, and public emergencies often generate visually similar footage. A video of an earlier flood, earthquake, protest, or fire may be presented as evidence of a new event. Since the footage itself is not necessarily manipulated, conventional deepfake detection methods cannot reliably identify the deception.

The central challenge is therefore not to determine whether the pixels are synthetic, but to determine whether the video has appeared before and whether the newly claimed context is compatible with its original publication history.

1.2 Research Motivation

Fact-checking organisations typically investigate resurfaced videos manually by extracting frames, performing reverse searches, comparing captions, checking earlier uploads, and tracing the most reliable source. This process is effective but time-consuming, especially during fast-moving events when many misleading posts may circulate simultaneously.

A Nepal-focused retrieval and verification pipeline can reduce the search space for investigators by automatically locating earlier occurrences of a query video and presenting structured evidence about its original date, location, event, and source. The project therefore combines multimedia processing, information retrieval, language processing, and evidence-based decision making in a practical system.

1.3 Scope

The project focuses on Nepal-related news videos published during the most recent three-year collection window available at the time of dataset creation. It does not at-

tempt to detect all misinformation, generate general truth judgments, or identify arbitrary deepfakes. Its scope is limited to determining whether a video in a new post matches an older video in the historical repository and whether the new post misrepresents the video's temporal or event context.

The system considers four input signals:

1. visual content from representative video frames;
2. speech content transcribed from the audio track;
3. on-screen text extracted from video frames;
4. post-level text and metadata, including caption and publication timestamp.

2. Problem Statement

Existing fake-news detection systems are primarily designed for text classification, manipulated-media detection, or broad multimodal veracity prediction. Such systems may fail when an authentic historical video is reposted under a false current caption because the video contains no visual forgery.

There is no readily available Nepal-focused system that maintains a searchable multimodal repository of recent historical news videos and uses that repository to identify resurfaced footage. As a result, verification remains largely manual and depends on the availability and experience of individual fact-checkers.

The research problem addressed in this project is therefore:

> *How can a multimodal system retrieve earlier occurrences of Nepal-related news videos from a three-year historical repository and determine whether a newly uploaded video is being falsely presented as recent?*

The problem contains two closely connected technical tasks:

a) robust retrieval of the same or partially reused footage despite cropping, compression, resizing, watermarks, subtitles, and short clip extraction;

b) evidence-based comparison between the claim made in the new post and the verified metadata of the retrieved historical video.

3. Objectives

3.1 Main Objective

To design and implement an evidence-grounded multimodal system that identifies whether a Nepal-related news video has appeared within a historical collection from the past three years and determines whether it has been reposted as a recent event.

3.2 Specific Objectives

1. To collect and organise Nepal-related news videos from verified public sources covering a three-year period.
2. To construct a structured dataset containing raw videos, publication metadata, captions, transcripts, OCR text, keyframes, perceptual hashes, and visual embeddings.
3. To extract representative visual, textual, and audio evidence from both historical and query videos.
4. To retrieve near-duplicate and partially reused videos using perceptual hashing, OpenCLIP embeddings, and FAISS-based similarity search.
5. To rerank retrieved candidates using visual, temporal, transcript, OCR, and source-reliability signals.
6. To compare the claim in the new post with the verified context of the best historical match.
7. To produce an interpretable decision with traceable supporting evidence.

4. Literature Review

4.1 Out-of-Context Multimedia Misinformation

Out-of-context misinformation uses genuine media together with false or misleading contextual information. The media may be authentic, but the accompanying claim changes the time, location, event, or identity associated with it. Research on multimodal misinformation has shown that image or video authenticity alone is insufficient for this task; verification requires comparison between the claimed context and external evidence.

Datasets such as NewsCLIPpings and VERITE demonstrate that semantically plausible but factually incorrect media-caption combinations are significantly harder to detect than random mismatches. These studies also show that models can exploit superficial shortcuts if evaluation splits are not carefully designed.

4.2 Video Fingerprinting and Near-Duplicate Retrieval

Perceptual hashing creates compact signatures that remain relatively stable under resizing, recompression, and minor visual changes. Hash-based matching is computationally efficient and suitable for detecting exact or near-exact copies. However, it is less reliable when videos are heavily cropped, watermarked, edited, or reduced to short segments.

Deep visual embeddings provide a more flexible representation. Contrastively pretrained encoders such as CLIP map images into a semantic feature space and can retrieve visually related frames even after moderate transformations. For video retrieval, frame-level features must also preserve temporal order so that a partial clip can be matched against a longer source video.

4.3 Large-Scale Similarity Search

FAISS provides efficient nearest-neighbour indexing for dense vectors. Instead of comparing a query embedding against every stored vector, FAISS uses optimised index structures to return a shortlist of likely candidates. This is suitable for searching a repository containing many videos and many keyframes per video.

4.4 Speech Transcription and OCR

Audio and on-screen text provide useful evidence when visual scenes are common or ambiguous. Whisper can transcribe multilingual and mixed-language speech, including Nepali and English content, with reasonable robustness to background noise. OCR can recover headlines, location names, dates, channel logos, and subtitles embedded in video frames. These signals can improve reranking and help reconstruct the video's original context.

4.5 Evidence-Grounded Verification

A similarity score alone does not establish that a post is misleading. The system must determine whether an earlier matching video exists, assess whether the source is credible, and compare the query claim with the earlier publication. Evidence grounding makes the decision traceable and reduces the risk of producing unsupported labels.

4.6 Research Gap

Existing multimodal misinformation benchmarks are mainly image-based or designed for general fake-news classification. Existing video-retrieval benchmarks focus on copy detection but do not verify whether a video is being reused with a false recent claim. Moreover, there is no focused benchmark or prototype designed around Nepal-related news videos, Nepali-English mixed text, and locally relevant event categories.

This project addresses that gap by combining video retrieval and contextual verification in a single Nepal-focused system.

5. Proposed Methodology

5.1 Research Design

The project follows an applied machine-learning and information-retrieval design. A historical multimedia repository is first constructed. The same preprocessing pipeline is then applied to each query video. Retrieval is performed in two stages: fast candidate generation followed by multimodal reranking. The final decision is rule-based or lightweight learned classification over evidence features rather than unrestricted generative reasoning.

5.2 End-to-End Workflow

Figure 5.1 presents the complete workflow.

5.3 Historical Dataset Construction

The dataset is limited to Nepal-related news videos published during a fixed three-year collection window. Candidate sources include verified YouTube channels and public archives of Nepali news organisations, public agencies, and fact-checking organisations. Only publicly accessible material with identifiable source and publication information is included.

The collection prioritises event categories in which resurfaced videos are common:

- floods, landslides, earthquakes, fires, and severe weather;
- protests, rallies, strikes, and political incidents;
- road accidents, aviation incidents, and infrastructure failures;
- public disorder, security incidents, and emergency response;
- major cultural, social, or national events.

For each video, the following fields are stored:

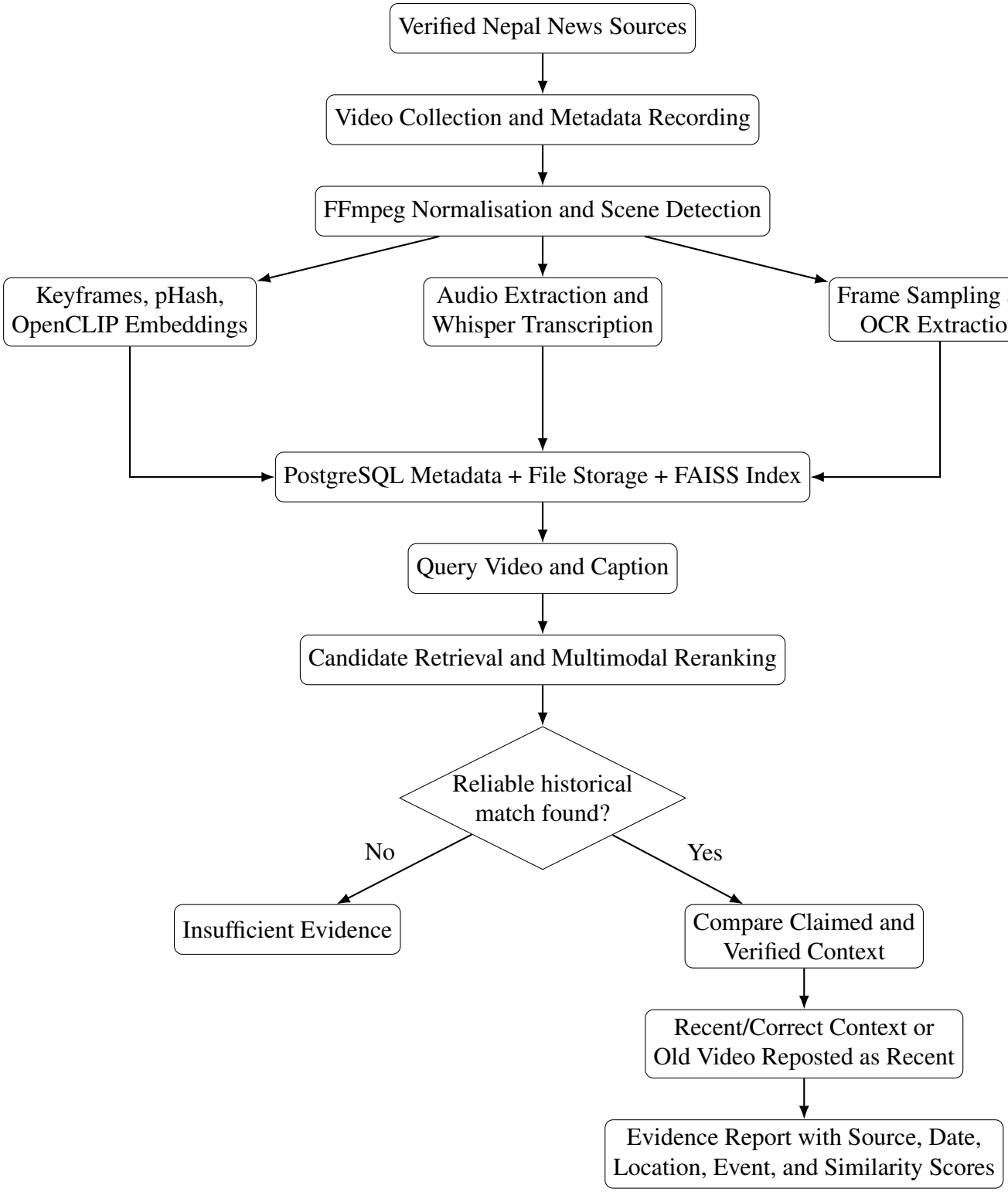


Figure 5.1: Proposed end-to-end workflow

Table 5.1: Dataset fields

Field	Description
Video identifier	Unique internal identifier
Source name	News organisation, agency, or verified channel
Source URL	Original public publication link
Publication timestamp	Date and time of the source upload
Title and caption	Original title, description, or accompanying text
Event category	Flood, protest, accident, fire, earthquake, etc.
Location	Province, district, municipality, or named place where available
Duration and format	Technical video properties
Transcript	Whisper-generated speech transcript
OCR text	Text extracted from representative frames
Keyframes	Scene-based or interval-based representative images
Perceptual hashes	Compact hashes for fast near-duplicate detection
Visual embeddings	OpenCLIP vectors for semantic retrieval
Evidence quality	Reliability score based on source and metadata completeness

5.4 Data Collection Procedure

The collection procedure is semi-automatic:

1. A crawler or source-specific downloader records public metadata and downloads permitted video files for research use.
2. Duplicate URLs and exact duplicate files are removed.
3. Videos are normalised to a standard container and frame rate using FFmpeg.
4. Metadata is reviewed and corrected manually where necessary.
5. Event category and location are annotated using a predefined guideline.
6. Videos without sufficient source information are excluded from the verified subset.

The system does not require an internet-scale collection. A focused and carefully verified repository is preferred over a much larger but unreliable dataset.

5.5 Video Preprocessing

Each video is processed using the following steps:

1. decode and normalise the video using FFmpeg;
2. detect scene boundaries using PySceneDetect;
3. select one or more representative frames per scene;
4. resize frames for embedding extraction while preserving aspect ratio;
5. extract the audio track as mono waveform;
6. sample frames for OCR;
7. save all derived artefacts using the unique video identifier.

A fallback interval-based keyframe strategy is used when scene detection returns too few or too many scenes.

5.6 Visual Feature Extraction

Two complementary visual representations are used.

5.6.1 Perceptual Hashing

Perceptual hashes are computed for keyframes. Hamming distance is used to identify exact or lightly transformed copies. This provides a fast first-pass signal and is especially effective against resizing, recompression, and minor colour changes.

5.6.2 OpenCLIP Embeddings

Each keyframe is encoded using a pretrained OpenCLIP image encoder. The resulting vectors are L2-normalised and stored in FAISS. A video-level candidate score is obtained by aggregating the best frame-to-frame similarities between the query and candidate video.

For a query frame embedding q_i and a candidate frame embedding c_j, cosine similarity is computed as:

$$S_{ij} = \frac{q_i \cdot c_j}{\|q_i\| \|c_j\|}. \tag{5.1}$$

A simple video-level visual score is:

$$S_{visual} = \frac{1}{N_q} \sum_{i=1}^{N_q} \max_j S_{ij}, \tag{5.2}$$

where N_q is the number of query keyframes.

5.7 Audio Transcription

The audio track is transcribed using Whisper. The transcript is stored with timestamps where available. Basic post-processing removes repeated filler tokens, normalises punctuation, and preserves mixed Nepali-English content. The transcript is not treated as ground truth; it is used as a supporting retrieval and verification signal.

Transcript similarity may be calculated using multilingual sentence embeddings or TF–IDF cosine similarity. Named locations, dates, organisations, and event words are extracted where possible.

5.8 Optical Character Recognition

OCR is applied to selected frames rather than every frame. The pipeline prioritises the first frames of scenes and frames containing stable text regions. Extracted text is cleaned and deduplicated. OCR may reveal channel names, place names, dates, headlines, and subtitles that are useful for candidate reranking.

5.9 Indexing and Storage

The prototype uses:

- PostgreSQL or SQLite for metadata and annotation records;
- local or object storage for original videos, audio, and keyframes;
- FAISS for dense visual-vector search;
- optional hash tables for perceptual-hash lookup.

Each FAISS vector stores a reference to its video identifier and frame timestamp. This allows the system to map retrieved frame matches back to full source videos.

5.10 Candidate Retrieval

Candidate retrieval is performed in two stages.

5.10.1 Stage 1: Fast Candidate Generation

The system searches perceptual hashes and FAISS visual embeddings. The top-K matching frames are grouped by source video. Candidate videos receiving matches from multiple query frames are prioritised.

5.10.2 Stage 2: Multimodal Reranking

Each candidate receives a combined score:

$$S_{final} = w_v S_{visual} + w_h S_{hash} + w_t S_{transcript} + w_o S_{ocr} + w_m S_{metadata}, \tag{5.3}$$

where the weights are tuned on a validation set and sum to one.

The reranking stage also checks whether the matched frames appear in a consistent temporal order. This reduces false matches caused by visually similar but unrelated scenes.

5.11 Claim Extraction

The query claim is created from:

- the social-media caption or user-provided text;
- OCR text extracted from the video;
- Whisper transcript;
- the publication timestamp of the query post.

The system extracts a compact structured claim:

$$C_q = \{d_q, l_q, e_q\}, \tag{5.4}$$

where d_q is the claimed date or recency, l_q is the location, and e_q is the event category.

Relative expressions such as *today*, *yesterday*, or *currently* are normalised using the query post timestamp.

5.12 Evidence Comparison and Decision Logic

The strongest historical candidate provides verified context:

$$C_h = \{d_h, l_h, e_h, s_h\}, \tag{5.5}$$

where s_h is the source record.

The system produces one of three outputs:

1. **Old video reposted as recent:** a reliable earlier match exists and the query presents it as a new or current event.
2. **Recent or correctly contextualised:** no contradictory earlier evidence is found, or the query accurately acknowledges the historical context.
3. **Insufficient evidence:** retrieval confidence or source quality is below the required threshold.

The prototype avoids making a stronger factual judgment than the evidence supports. In particular, absence of a match does not prove that a video is new; it only means that no sufficiently reliable match was found in the available repository.

5.13 Evidence Report

The final report contains:

- decision label;
- confidence score;
- best matching historical video;
- original publication date and source;
- matched frame timestamps;
- extracted query claim;
- short explanation of the detected mismatch;
- warning when evidence is incomplete.

6. System Architecture and Implementation Plan

6.1 Architecture

The prototype follows a modular architecture shown in Figure 6.1.

6.2 Technology Stack

6.3 Development Phases

The work is organised into the following implementation phases without requiring all modules to be developed simultaneously:

1. finalise annotation guidelines and source list;
2. collect and clean the historical video repository;
3. implement keyframe, Whisper, OCR, pHash, and OpenCLIP preprocessing;
4. build FAISS and metadata indexes;
5. implement candidate grouping and reranking;
6. implement claim extraction and decision logic;
7. integrate the API and interface;
8. evaluate retrieval, classification, robustness, and explanation quality;
9. document limitations and prepare the final prototype.

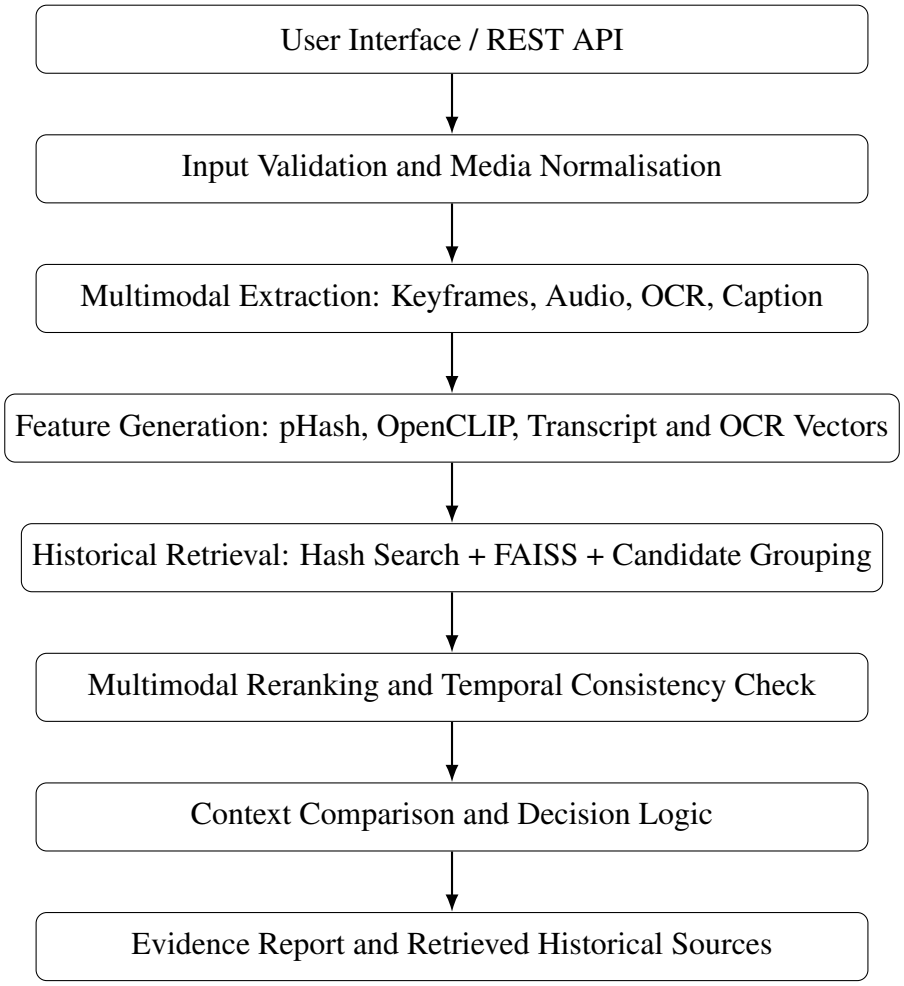


Figure 6.1: High-level system architecture

Table 6.1: Proposed technology stack

Component	**Technology**
Programming language	Python
Video processing	FFmpeg, OpenCV, PySceneDetect
Speech recognition	Whisper
OCR	EasyOCR or PaddleOCR
Visual embeddings	OpenCLIP
Perceptual hashing	ImageHash or VideoHash
Vector search	FAISS
NLP utilities	Hugging Face Transformers, dateparser, spaCy or rule-based extraction
Metadata database	PostgreSQL or SQLite
Backend	FastAPI
Experiment tracking	CSV/JSON logs or MLflow
Deployment	Docker

7. Evaluation Plan

7.1 Dataset Splitting

All reposted or transformed variants derived from the same source video are kept in the same split. This source-video-disjoint strategy prevents the model from seeing one version of a video during training and another version during testing.

Where the dataset size permits, an event-disjoint split is additionally used so that videos from the same incident are not distributed across both training and testing sets.

7.2 Retrieval Evaluation

The retrieval component is evaluated using:

- Recall@K;
- mean Average Precision (mAP);
- Mean Reciprocal Rank (MRR);
- top-1 retrieval accuracy;
- average query latency.

A query is considered correctly retrieved when the original source video or a verified earlier occurrence appears within the top-K candidates.

7.3 Decision Evaluation

The final three-class decision is evaluated using:

- accuracy;
- precision, recall, and F1-score for each class;
- macro-F1 to account for class imbalance;
- confusion matrix;
- abstention coverage and selective accuracy.

7.4 Robustness Tests

Query videos are transformed to reflect common social-media modifications:

- recompression;
- cropping;
- resizing;
- watermark insertion;
- subtitle overlay;
- horizontal flipping;
- speed adjustment;
- partial clip extraction;
- audio removal or replacement.

Performance before and after transformation is compared.

7.5 Ablation Study

The following ablations are used to measure the contribution of each modality:

1. OpenCLIP only;
2. perceptual hash only;
3. visual features without OCR;
4. visual features without Whisper transcript;
5. no temporal consistency check;
6. no metadata/source-reliability score;
7. no abstention threshold.

7.6 Human Evaluation of Evidence Reports

A small set of reports is reviewed by project members and, where possible, journalism or fact-checking reviewers. Reports are rated for:

- correctness of the retrieved source;
- relevance of the explanation;
- faithfulness to evidence;
- clarity for a non-technical user.

8. Feasibility, Challenges, and Mitigation

8.1 Feasibility

The project relies mainly on pretrained models and efficient indexing rather than training a large multimodal model from scratch. OpenCLIP and Whisper can be used directly or with limited adaptation. FAISS supports scalable retrieval, and the system can initially be evaluated on a carefully selected repository rather than an internet-scale archive.

8.2 Expected Challenges

- incomplete or deleted source videos;
- low-quality uploads and heavy recompression;
- short clips extracted from long broadcasts;
- weak Nepali OCR accuracy;
- mixed Nepali-English speech and text;
- visually similar but unrelated disaster footage;
- unreliable captions and missing location metadata;
- limited compute and storage resources.

8.3 Mitigation Strategies

- combine hash and embedding retrieval rather than relying on one signal;
- use multiple keyframes per video;
- include temporal consistency in reranking;
- treat OCR and transcript as supporting evidence rather than absolute truth;
- manually verify the gold evaluation subset;
- introduce a source-reliability score;
- use an explicit insufficient-evidence class;

- use pretrained lightweight models and batch preprocessing;
- store compressed video copies and derived features separately.

8.4 Ethical and Legal Considerations

Only publicly accessible videos are collected, and source links are retained for attribution. The dataset is used for academic research and verification rather than content redistribution. The system does not accuse individual users of intentionally spreading misinformation; it reports evidence about the relationship between a query post and an earlier publication. Outputs are presented as decision support rather than unquestionable truth.

9. Expected Outcomes

The expected outcomes are:

1. a curated Nepal-focused historical news-video repository covering three years;
2. a reproducible preprocessing pipeline for keyframes, Whisper transcripts, OCR text, perceptual hashes, and OpenCLIP embeddings;
3. a FAISS-based retrieval system for earlier occurrences of a query video;
4. a multimodal reranking method robust to common social-media transformations;
5. a three-class evidence-grounded decision module;
6. a prototype web or API interface that returns matched sources and explanations;
7. an evaluation report covering retrieval accuracy, decision performance, robustness, and ablation results.

The project contributes a focused, locally relevant approach to a misinformation problem that is not adequately addressed by deepfake detection or text-only fake-news classification. Its primary contribution is the integration of provenance retrieval and contextual verification for Nepal-related news videos.

10. Conclusion

This proposal presents a practical system for detecting authentic old videos that are reposted as recent Nepal-related news. The project narrows the problem to a searchable three-year historical repository and uses a feasible combination of established tools: FFmpeg and OpenCV for video processing, PySceneDetect for keyframes, Whisper for transcripts, OCR for on-screen text, OpenCLIP for visual representation, and FAISS for retrieval.

The system is intentionally evidence-grounded. It does not classify a post solely from learned appearance patterns; instead, it retrieves earlier occurrences, checks source records, compares contextual claims, and reports the supporting evidence. This design makes the system more interpretable, more useful to fact-checkers, and more realistic for an undergraduate major project.

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